

1. What is Artificial Intelligence (AI)?

Definition

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence.

These tasks include:

- Learning
 - Reasoning
 - Problem Solving
 - Decision Making
 - Language Understanding
 - Image Recognition
-

Examples of AI

Chatbots

Example:

- ChatGPT
 - Customer Support Bots
-

Face Recognition

Example:

- Mobile Face Unlock
-

Recommendation Systems

Example:

- Netflix Recommendations
 - YouTube Suggestions
-

Self-Driving Cars

AI helps vehicles:

- Detect roads
 - Identify pedestrians
 - Make driving decisions
-

Real-Life Example

Suppose a human sees:

□

Human immediately says:

Dog

AI can be trained to do the same.

2. Types of AI

Narrow AI

Designed for a specific task.

Examples:

- Siri
- Alexa
- Google Assistant

Most AI today is Narrow AI.

General AI (AGI)

Can perform any intellectual task a human can perform.

Examples:

Currently does not exist.

Super AI

Would exceed human intelligence.

Still theoretical.

3. What is Machine Learning?

Definition

Machine Learning (ML) is a subset of AI where machines learn patterns from data without being explicitly programmed.

Traditional Programming

Input + Rules
↓
Output

Example:

```
marks = 70

if marks >= 40:
    print("Pass")
else:
    print("Fail")
```

Rules are manually written.

Machine Learning

Input + Output Data
↓
Machine Learns Rules
↓
Prediction

Example:

Student Data:

Study Hours Marks

Study Hours Marks

2	40
4	60
6	80

Machine learns relationship.

ML Example

House Price Prediction

Input:

Area
Bedrooms
Location

Output:

House Price

4. Types of Machine Learning

Supervised Learning

Uses labeled data.

Example:

Email → Spam
Email → Not Spam

Algorithms:

- Linear Regression
 - Logistic Regression
 - Random Forest
 - KNN
-

Unsupervised Learning

Uses unlabeled data.

Example:

Customer Segmentation.

Algorithms:

- K-Means
 - Hierarchical Clustering
-

Reinforcement Learning

Agent learns through rewards and penalties.

Example:

- Game Playing
 - Robotics
-

5. Limitations of Traditional Machine Learning

Machine Learning works well for:

- Small datasets
- Structured data

But struggles with:

- Images
 - Audio
 - Videos
 - Natural Language
-

Example: Image Recognition

Suppose we want to identify a cat.

Traditional ML requires:

Feature Extraction

Manually creating features like:

- Ear Shape
- Tail Length
- Fur Pattern

This is difficult.

Problem

Feature engineering becomes:

Time Consuming

and

Complex

6. What is Deep Learning?

Definition

Deep Learning is a subset of Machine Learning that uses Artificial Neural Networks with multiple hidden layers to automatically learn patterns from data.

Relationship

Artificial Intelligence

↓

Machine Learning

↓

Deep Learning

Why Called "Deep"?

Because neural networks contain:

Multiple Hidden Layers

Example:

Input Layer
↓
Hidden Layer 1
↓
Hidden Layer 2
↓
Hidden Layer 3
↓
Output Layer

The greater number of hidden layers makes the network "deep."

7. Deep Learning vs Machine Learning

Machine Learning	Deep Learning
Requires feature engineering	Learns features automatically
Works on smaller data	Needs large data
Faster training	Slower training
Less computation	More computation
Easier interpretation	Harder interpretation

Example

ML Approach

Cat Detection:

Engineer manually extracts:

Ear Shape
Tail Length
Color

DL Approach

Provide image.

Network learns features automatically.

8. Why Deep Learning Became Popular

Three reasons:

1. Large Amounts of Data

Examples:

- Social Media
- Images
- Videos
- Sensors

2. Powerful GPUs

GPUs perform parallel computations.

Examples:

- NVIDIA GPUs

Training became much faster.

3. Improved Algorithms

Examples:

- CNN
- RNN
- LSTM
- Transformers

9. Real-World Applications of Deep Learning

Computer Vision

Teaching computers to understand images.

Applications:

- Face Recognition
 - Object Detection
 - Medical Imaging
-

Example

Upload image:

Dog Image

Output:

Dog

Natural Language Processing (NLP)

Teaching computers to understand text.

Applications:

- ChatGPT
 - Translation
 - Summarization
-

Example

Input:

Translate English to Hindi

Output:

नमस्ते

Speech Recognition

Applications:

- Google Assistant
- Siri

Example:

Speech



Text

Autonomous Vehicles

Self-driving cars use DL for:

- Lane Detection
 - Traffic Sign Recognition
 - Pedestrian Detection
-

Healthcare

Applications:

- Disease Detection
 - Medical Imaging
 - Cancer Detection
-

10. Famous Deep Learning Models

ANN (Artificial Neural Network)

Basic neural network.

Used for:

- Classification
 - Regression
-

CNN (Convolutional Neural Network)

Used for images.

Applications:

- Face Recognition
 - Object Detection
-

RNN (Recurrent Neural Network)

Used for sequence data.

Applications:

- Language Modeling
 - Text Generation
-

LSTM

Improved RNN.

Handles long-term memory.

Applications:

- Translation
 - Chatbots
-

Transformers

Most modern architecture.

Used in:

- ChatGPT
 - Gemini
 - Claude
-

11. Deep Learning Workflow

```
Data Collection
  ↓
Data Preprocessing
  ↓
Neural Network
  ↓
```

Training
↓
Evaluation
↓
Prediction

Example

House Price Prediction

Dataset
↓
Train Model
↓
Evaluate
↓
Predict Price

12. Advantages of Deep Learning

Automatic Feature Learning

No manual feature engineering.

High Accuracy

Often outperforms traditional ML.

Handles Large Data

Works well with:

- Images
 - Videos
 - Audio
-

End-to-End Learning

Input:

Image

Output:

Prediction

No manual feature extraction required.

13. Limitations of Deep Learning

Requires Large Data

Small datasets may not work well.

Computationally Expensive

Requires:

- GPU
 - Memory
 - Processing Power
-

Longer Training Time

Training can take:

- Hours
 - Days
 - Weeks
-

Black Box Nature

Difficult to explain decisions.

14. Popular Deep Learning Frameworks

TensorFlow

Created by Google.

```
import tensorflow as tf
```

Keras

High-level API for TensorFlow.

```
from tensorflow import keras
```

PyTorch

Created by Meta.

```
import torch
```

15. Mini Deep Learning Example

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

model = Sequential()

model.add(Dense(5, input_shape=(3,), activation='relu'))

model.add(Dense(1))

model.summary()
```

Explanation

```
Input Layer → 3 Features
Hidden Layer → 5 Neurons
Output Layer → 1 Neuron
```

Interview Questions

What is Deep Learning?

Deep Learning is a subset of Machine Learning that uses neural networks with multiple hidden layers to learn complex patterns automatically.

Why is Deep Learning called "Deep"?

Because it contains multiple hidden layers between input and output layers.

Difference between ML and DL?

ML requires manual feature engineering, while DL automatically learns features from data.

Why is Deep Learning popular?

- Large datasets
 - Powerful GPUs
 - Improved algorithms
-

Name some Deep Learning architectures.

- ANN
- CNN
- RNN
- LSTM
- Transformers